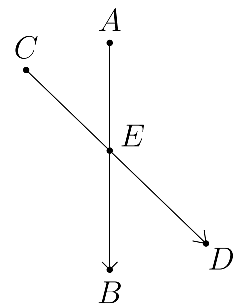
Combinatorial Geometry — Solutions

Problem 1. (JBMO 2007 Problem 3) Given are 50 points in the plane, no three of them belonging to a same line. Each of these points is coloured using one of four given colours. Prove that there is a colour and at least 130 scalene triangles with vertices of that colour.

Solution. By the pigeonhole principle, there is a colour such that at least $\left\lceil \frac{50}{4} \right\rceil = 13$ points are of that colour. It remains to show that for any 13 points, we can form at least 130 scalene triangles using them as vertices. There are $\binom{13}{3} = 286$ triangles in total. For each pair of points A and B, there are at most two points on the perpendicular bisector of AB since no three points are collinear. Therefore, AB is the base of at most two isosceles triangles. As a result, there are at most $2\binom{13}{2} = 156$ isosceles triangles. (Note that each equilateral triangle is counted three times, but this does not affect the argument.) Thus, we see that there are at least $286 - 156 = 130$ scalene triangles. □

Problem 2. (239 Open MO 2018 Senior Problem 1) Prove that in any convex polygon where all pairwise distances between vertices are distinct, there exists a vertex such that the closest vertex of the polygon is adjacent to it.

Solution.



For each vertex P, we let $f(P)$ be the vertex which is closest to P. Suppose on the contrary that there is no P such that P and $f(P)$ are adjacent.

Choose a vertex A such that A and $B = f(A)$ are separated by the smallest number of vertices. Let C be the vertex adjacent to A that lies on the side of the line AB with fewer vertices. By the assumption, C is distinct from A and B, and $D = f(C)$ should not lie on the same side of the line AB as C by the choice of A. Therefore, the segments AB and CD intersect at some point E. Note that $AD > AB$ and $BC > CD$ since $B = f(A)$ and $D = f(C)$. By the triangle inequality, we have

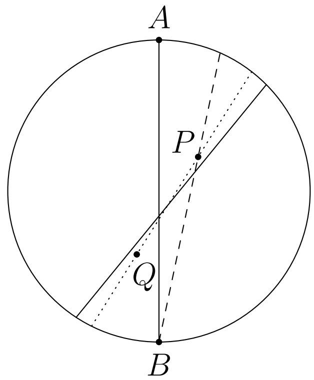
$$AB + CD = (AE + DE) + (BE + CE) > AD + BC > AB + CD.$$

This is a contradiction. Therefore, there must be a vertex P such that P and $f(P)$ are adjacent. □

Problem 3. There are $2m$ red points and $2n$ blue points on the plane such that no three points are collinear. Prove that there exists a line that separates all points into two halves, such that each half contains exactly m red points and n blue points.

Solution. We draw a circle that includes all points inside. It is possible to find a line ℓ that separates all points into two equal halves (which means each half contains exactly $m+n$ points). For example, we can start with a tangent to the circle which is not parallel to any line joining two of the given points. Then we translate the line across the circle in a parallel way. There must be a moment when it has crossed exactly $m+n$ points. We now rotate the line ℓ (clockwise) in the following way.

Let ℓ cut the circle at A and B. Firstly, we rotate ℓ about B until it passes through a coloured point P. Then we rotate ℓ about P until it meets another coloured point Q. Note that P and Q are separated by ℓ before this process. Thus, if we further rotate ℓ about the midpoint of PQ by a sufficiently small angle, the points P and Q will move to the opposite region separated by ℓ at the same time, and all other points remain in their original regions. This means each side of ℓ still contains exactly $m+n$ points.



Repeating the same process, eventually ℓ must have rotated by $180°$. The line ℓ at this stage must be parallel to the initial line AB. Clearly, there cannot be any point between ℓ and AB. Thus, we may further translate ℓ to AB without affecting the relative position of the points.

Throughout the whole process, we can identify two regions I and II separated by ℓ so that the two regions change in a continuous way. Let N be the number of red points in region I minus the number of red points in region II. Note that N must be even as there is an even number of red points in total. If the initial value of N is N_0, then its value becomes $-N_0$ at the end of the above process since regions I and II are swapped. As the value of N only changes by $0, \pm 2$ every time when ℓ crosses two points, there must be a moment when N is equal to 0. At this moment, there are m red points on each side of ℓ. Since there are $m+n$ points on each side of ℓ, there are exactly n blue points on each side. This is the desired straight line. $\square$

Problem 4. There is a convex polygon $\mathcal{P}$ on the coordinate plane. It is known that no matter how we translate $\mathcal{P}$ on the plane (without rotating and reflecting it), there is always a lattice point (a point with integer coordinates) covered by $\mathcal{P}$. Prove that there are two points on or inside $\mathcal{P}$ whose distance is at least $\sqrt{2}$.

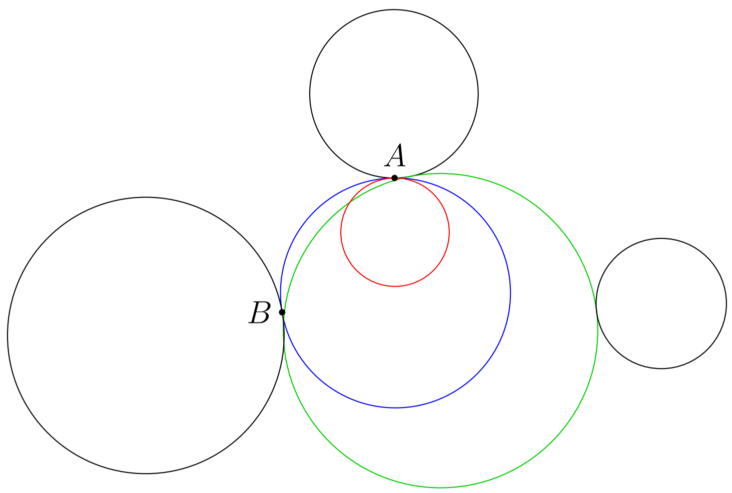
Solution. Suppose on the contrary that every two points on or inside $\mathcal{P}$ have distance less than $\sqrt{2}$. Translate $\mathcal{P}$ such that the origin O lies on the boundary of $\mathcal{P}$. If there is no other lattice point covered by $\mathcal{P}$, we can move $\mathcal{P}$ slightly such that O is no longer covered by $\mathcal{P}$ while there is no lattice point covered by $\mathcal{P}$. This contradicts the given condition. Therefore, there is another lattice point A covered by $\mathcal{P}$.

We are done if $OA \geqslant \sqrt{2}$. Otherwise, A must be one of $(1,0)$, $(0,1)$, $(-1,0)$ and $(0,-1)$. If two of these points are covered by $\mathcal{P}$, then we are also done because the distance between any two of them is at least $\sqrt{2}$. Therefore, A is uniquely determined. We can define a function f as follows. For every point X on the boundary of $\mathcal{P}$, if we translate $\mathcal{P}$ to the position satisfying $O = X$, then $f(X)$ is the unique point A defined above. It is clear that f is a continuous function since moving X continuously on the boundary of $\mathcal{P}$ means we continuously move $\mathcal{P}$ on the plane. As the codomain of f only consists of four distinct points, f must be a constant function.

WLOG assume $f(X) = (1,0)$ for all X. However, when X is one of the rightmost points of $\mathcal{P}$, it is clear that $(1,0)$ cannot be covered by P when $X = O$. This contradicts $f(X) = (1,0)$, and so we are done. $\square$

Problem 5. (USAMO 2007 Problem 2) A square grid on the Euclidean plane consists of all points (m,n), where m and n are integers. Is it possible to cover all grid points by an infinite family of discs with non-overlapping interiors if each disc in the family has radius at least 5?

Solution. No. Clearly, the discs cannot cover the whole plane. Choose any point circle which is not covered by any disc. We enlarge this circle about its centre until it touches a disc at some point A. Then we enlarge this circle about A until it touches a second disc at B. Lastly, we enlarge this circle such that it is always tangent to the first two discs until it touches a third disc (for example, by moving its centre away from the line AB along the perpendicular bisector of AB).



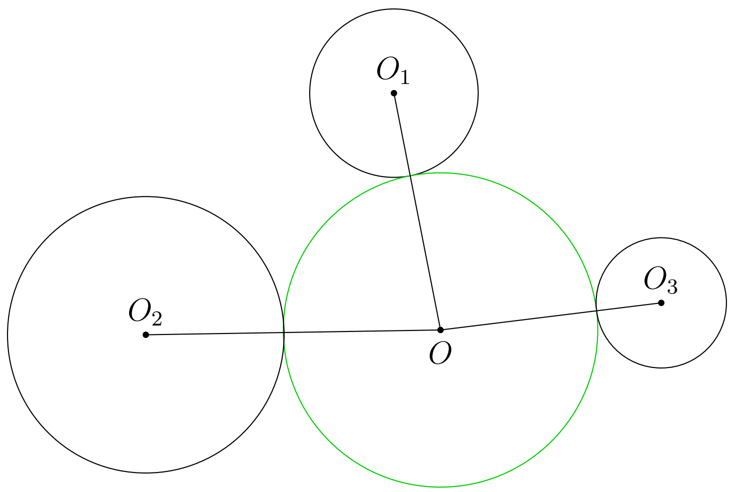
Now, we have obtained a circle with centre O and radius r which is tangent to three discs with centres O_1, O_2, O_3 and radii r_1, r_2, r_3 respectively. No matter O lies inside $\triangle O_1O_2O_3$ or not, we can always assume $\angle O_1OO_2 \leqslant 120^\circ$ up to symmetry. Applying the cosine formula to $\triangle OO_1O_2$, we have

$$\begin{aligned}(r_1+r_2)^2 &= (r+r_1)^2 + (r+r_2)^2 - 2(r+r_1)(r+r_2)\cos\angle O_1OO_2 \\ &\leqslant (r+r_1)^2 + (r+r_2)^2 + (r+r_1)(r+r_2).\end{aligned}$$

This is the same as $r_1r_2 \leqslant 3r^2 + 3(r_1+r_2)r$. It follows that

$$1 \leqslant 3r\left(\frac{r}{r_1r_2} + \frac{1}{r_1} + \frac{1}{r_2}\right) \leqslant 3r\left(\frac{r}{5^2} + \frac{1}{5} + \frac{1}{5}\right),$$

or $3r^2 + 30r - 25 \geqslant 0$. This implies $r \geqslant \dfrac{-15+10\sqrt{3}}{3}$.



There is a square enclosed by the circle with side length $\sqrt{2}r \geqslant \dfrac{-15\sqrt{2}+10\sqrt{6}}{3} > 1$. Thus, there must be a lattice point inside the square, and clearly this lattice point is not covered by any disc by the construction of the circle. $\square$